Prasiddha Thapa

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Education

Howard University – COAS Honors Program (Presidential Scholar) Expected graduation: 2030
B.S. in Mathematics

Current coursework: Proof and Problem Solving; Introduction to Data Science; Mathematics Research I

Honors

67th International Mathematical Olympiad — Honorable Mention 2026
Represented Nepal; joint-highest score on the national team.

4th International Mathematics Summer Camp — Bronze Medal 2026
Individual competition; highest-scoring participant from Nepal. Beijing, China.

Research & Computational Projects

Temporal Evidence Accumulation in Spiking Neural Networks 2026
Research under Prof. Christopher Kim, Howard University

- Built a **PyTorch spiking RNN** with Dale-constrained 80:20 excitatory/inhibitory connectivity and surrogate-gradient training.
- Conducted circuit ablations on a noisy decision task: **86.8% accuracy** (full model) vs. **63.9%** (no recurrence) and **48.1%** (no inhibition).
- Demonstrated temporal evidence integration: accuracy improved from **69% to 93%** as stimulus duration increased from 100 to 800 ms.

Grokking: Delayed Generalization in Neural Networks 2026

- Trained a **multilayer perceptron (MLP)** on modular addition over $\mathbb{Z}_{113}$, using one-hot encoding and a 30:70 train-test split.
- Investigated **delayed generalization (grokking)** over 17,000 training epochs, analyzing training and test accuracy to study the transition from memorization to generalization.

Household Pet-Readiness Modeling 2024
International Mathematical Modeling Challenge — Honorable Mention, Team

- Led mathematical modeling and data collection; implemented a Python readiness index combining financial, healthcare, and environmental factors using normalization and weighted scoring.
- Applied the model to demographic and economic data from the United States, Germany, and Canada; adapted its assumptions for different pet species and multi-pet households.

Mathematical Teaching & Leadership

Writer — Olympiad Mathematics Handouts

- Co-authored and typeset preparation material on algebraic inequalities and number theory to prepare handouts in LaTeX under Prajit Adhikari, a former IMO participant, for the math olympiad participants.

Founder & Captain — Mathematics Olympiad Team

- Led preparation sessions and captained teams representing St. Xavier's College; also mentored beginners through Nepal's national mathematics Discord community.

Technical and Research Skills

Programming and Tools: Python, PyTorch, NumPy, scikit-learn, Matplotlib, Git, Jupyter Notebook
Machine Learning: Deep Learning, Attention Mechanisms, Recurrent Neural Networks, Spiking Neural Networks, Multi-Task Learning, Sequence Modeling
Research Methods: Experimental Design, Model Evaluation, Ablation Studies, Monte Carlo Simulation, Numerical Optimization
Mathematics: Combinatorics, Discrete Mathematics, Probability, Mathematical Proofs